this process involves the purification of the desired product.

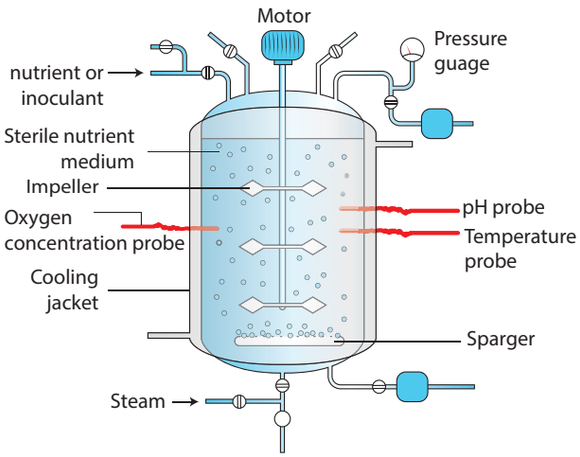


Figure 4.2: Bioreactor

Procedure of Fermentation

a. Depending upon the type of product, bioreactor is selected.

b. A suitable substrate in liquid medium is added at a specific temperature, pH and then diluted.

c. The organism (microbe, animal/plant cell, sub-cellular organelle or enzyme) is added to it.

d. Then it is incubated at a specific temperature for the specified time.

e. The incubation may either be aerobic or anaerobic.

f. Withdrawal of product using downstream processing methods

Application of fermentation in industries

Fermentation has industrial application such as:

1. Microbial biomass production

Microbial cells (biomass) like algae, bacteria, yeast, fungi are grown, dried and used as source of a complete protein called 'single cell protein (SCP)' which serves as human food or animal feed.

2. Microbial metabolites

Microbes produce compounds that are very useful to man and animals. These compounds called metabolites, can be grouped into two categories:

a. **Primary metabolites:** Metabolites produced for the maintenance of life process of microbes are known as primary metabolites Eg. Ethanol, citric, acid, lactic acid, acetic acid.

b. **Secondary metabolites:** Secondary metabolites are those which are not required for the vital life process of microbes, but have value added nature, this includes antibiotics e.g -Amphotericin-B (*Streptomyces nodosus*), Penicillin (*Penicillium chryosogenum*) Streptomycin (*S. grises*) , Tetracycline (*S. aureofacins*), alkaloids, toxic pigments, vitamins etc.

3. Microbial enzymes

When microbes are cultured, they secrete some enzymes into the growth media. These enzymes are industrially used in detergents, food processing, brewing and pharmaceuticals. Eg. protease, amylase, isomerase, and lipase.

4. Bioconversion, biotransformation or modification of the substrate

The fermenting microbes have the capacity to produce valuable products, eg. conversion of ethanol to acetic acid (vinegar), isopropanol to acetone, sorbitol to sorbose (this is used in the manufacture of vitamin C), sterols to steroids.

4.3.2 Single Cell Protein (SCP)

Single cell proteins are dried cells of microorganism that are used as protein supplement in human foods or animal feeds. Single Cell Protein (SCP) offers an unconventional but plausible solution to protein deficiency faced by the entire humanity. Although single cell protein has high nutritive value due to its higher protein, vitamin, essential amino acids and lipid content, there are doubts on whether it could replace conventional protein sources due to its high nucleic acid content and slower

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